

Class Notes: Phylum Echinodermata

Animal Kingdom • PU-I Biology • Class Notes

Body Plan and Water Vascular System

Echinodermata means spiny-bodied due to an endoskeleton of calcareous ossicles; exclusively marine; bilaterally symmetrical, triploblastic, coelomate, organ system level. Digestive system is complete with mouth on the ventral side and anus on the dorsal side.

Reproduction and Examples

Water vascular / ambulacral system is a distinctive feature, helping in locomotion, food capture/transport and respiration. Reproduction is sexual with external fertilisation and indirect development (Bipinnaria larva).

Examples: Asterias (star fish), Echinus (sea urchin), Cucumaria (sea cucumber), Ophiura (brittle star), Antedon (sea lily).